Spurious Precision:
Why Shift-Share Designs Overstate Monetary Policy Effects on Local Employment

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Plain-Language Summary

What was asked. When the Federal Reserve raises interest rates, some local economies should suffer more than others. Places built on construction and factories borrow more and build more. A standard method finds that these places actually do *better* when rates rise, and finds it with great confidence. This paper asks whether that confidence is real.

What was found. It is not. The method measures policy using the interest rate the Fed actually set. But the Fed raises rates when the economy is already strong, so that measure carries the economy inside it. Swapping in a measure of only the surprise part of policy leaves the estimates far too uncertain to say anything. The margin of error grows by roughly five to twelve times, depending on the sample. Separately, the method fails its own built-in check for whether the comparison was fair to begin with.

What it means. The tight margins in this kind of study can come from economic ups and downs rather than from policy. Researchers using this design should report both measures side by side and compare the margins of error, not just the estimates. A large gap between them is a warning.

The honest limitation. This paper shows the original confidence was misplaced. It does not recover the true effect. The cleaner estimates are too imprecise to say whether employment rises or falls, and one further test of the margins of error remains undone.

1 Introduction

Monetary policy does not land evenly across a country. A rate rise raises the cost of borrowing, and places whose employment sits in interest-sensitive industries should feel it first. Construction and durable manufacturing are the usual examples. Carlino and DeFina (1998) established the regional version of this claim, and it has been standard since.

*DePaul University. Email: `dalisabdallah@gmail.com`. Replication code: `causal-did/Dalis_Abdallah_RebuildExposure.R` and the scripts it feeds. I thank Prof. Jin Man Lee for advising this research program. All errors are my own.

The workhorse tool for testing it is the shift-share, or Bartik, design (Bartik, 1991). A national shock is interacted with a local exposure share fixed before the sample begins. Variation in exposure does the identifying work, and the fixed share is meant to keep exposure from responding to the shock (Goldsmith-Pinkham et al., 2020).

The design has a measurement problem at its centre, and it is not a subtle one. Applied work usually measures policy with the realized change in the federal funds rate. That change is endogenous. The Federal Reserve tightens when the economy runs hot, and the sectors that define exposure are the most cyclical ones in the economy.

Everyone writing in this literature knows that. The high-frequency identification programme exists because of it (Nakamura and Steinsson, 2018; Bauer and Swanson, 2023). What is missing is a price. How much does the endogenous measure distort a local labor market design, and along which dimension does the distortion show up?

The instinctive answer is bias, meaning the coefficient sits in the wrong place. This paper finds something different, and more awkward.

I estimate a shift-share difference-in-differences design on county employment from 2002 to 2019, using the predetermined 2002 county share of employment in construction and manufacturing as exposure. Every result appears for the full window and again excluding 2009 through 2012.

Three findings follow.

First, the naive design looks strong and points the wrong way. The contemporaneous interaction is 0.033 with a p-value below 0.001. Read directly, tightening raises employment where exposure is highest.

Second, the design fails its own falsification test. Adding leads of the interaction, two of the four reject zero under both clustered and randomization inference. Employment in exposed counties was already moving before policy changed.

Third, and this is the paper's central result, replacing the realized rate change with Bauer–Swanson high-frequency surprises does not relocate the estimate. It dissolves it. No horizon rejects zero, in either window, and standard errors widen by a factor of five to ten. The two estimators are not statistically distinguishable at any horizon.

The mechanism is visible in the shock series themselves. Over this sample the realized rate change correlates with contemporaneous national employment growth at 0.470. The identified surprise correlates at -0.119, and the two measures correlate with each other at -0.010.

So the naive design draws its precision from business-cycle variation rather than policy variation. Cyclical variation is abundant and moves with the outcome, which is what produces tight intervals. It is also the variation an identification strategy exists to remove. The design was not estimating the effect of monetary policy precisely. It was estimating something else precisely.

The contribution is diagnostic rather than structural. This paper does not propose a new estimator, and it does not recover the true employment response. It shows that a widely used design can report a precise and interpretable coefficient whose precision is an artifact, and that two cheap checks catch it.

The recommendation follows from that and costs one column. Report the design under both shock measures, and read the comparison on the standard errors rather than the coefficients. A large gap says the reported precision was borrowed.

This is a complement to the shift-share inference literature, not a substitute for it. Adão et al. (2019) show that conventional standard errors can be severely understated when a small number of common sectoral shocks drive exposure. That concern is live here and is not tested in this paper. The problem documented here is upstream of it: before asking whether the standard errors are computed correctly, one may ask what variation they are computed on.

Section 2 describes the data and both designs. Section 3 reports the results, figure by figure. Section 4 discusses what the precision was made of, what the two diagnostics do and do not establish, and what remains untested.

2 Data and Method

3 Data and Methods

3.1 Data

Four sources are used, all public.

County employment comes from the Quarterly Census of Employment and Wages, assembled as a county-by-quarter panel covering 2002 through 2019. The outcome is the log of county employment. Observations with non-positive employment are dropped.

The screen is applied to the employment level rather than to its logarithm. The panel records log employment as zero in the small number of county-quarters where employment itself is zero, so a finiteness check on the log variable would retain them. Those observations sit roughly nine log points below the median county and are concentrated in very small counties and at county boundary-recodes; they are recording artifacts rather than economic zeros.

Exposure comes from County Business Patterns. It is the share of county employment in construction and manufacturing in 2002, the first year of the panel. The share is fixed at its 2002 value for every quarter, so it cannot respond to later policy.

The Quarterly Census of Employment and Wages panel covers 3,228 counties. The County Business Patterns exposure file covers 3,127. Joining them, and requiring an observed funds-rate change, leaves **3,127 counties** over 72 quarters, and that is the estimation sample in both windows.

The 2002 County Business Patterns county file records the District of Columbia as a statewide entry rather than a county, and it is the only jurisdiction it treats that way. It is recoded here to its standard county FIPS and retained; the alternative, dropping every statewide entry, would silently exclude a real jurisdiction along with the fifty genuine residual records. The District has the lowest exposure in the country at 0.020.

Attrition beyond that is mechanical. The joined panel has 224,952 county-quarter observations in the full window and 174,968 excluding 2009 through 2012. The main specification uses all of them. The specification that adds four pre-period leads uses 212,468 and 162,484 respectively, the difference being rows whose leads are not observed. Counts reported in the paper are estimation samples unless stated otherwise.

Macroeconomic series come from FRED: the effective federal funds rate, total nonfarm payrolls, and the consumer price index. The funds rate is averaged within quarter and first-differenced.

The identified policy shock is the orthogonalized Bauer–Swanson monetary policy surprise, taken from the Federal Reserve Bank of San Francisco's published workbook, sheet "Monthly (update 2023)". Monthly surprises are summed within quarter. The series is constructed to be orthogonal to macroeconomic and financial information available before each meeting.

3.2 The shift-share difference-in-differences design

The main specification is

$$\ln E_{ct} \;=\; \beta\left(\Delta f_t \times s_c\right) \;+\; \alpha_c \;+\; \tau_t \;+\; \varepsilon_{ct}, \tag{1}$$

where $\ln E_{ct}$ is log employment in county c in quarter t, Δf_t is the quarterly change in the federal funds rate, and s_c is the 2002 exposure share. County fixed effects α_c absorb permanent differences in level, and quarter fixed effects τ_t absorb every national shock, including Δf_t itself.

The funds rate change therefore does not enter separately. It is collinear with the quarter fixed effects and is absorbed by them. Only the interaction is identified, and β is a difference in differences: the extra employment response of a more exposed county relative to a less exposed one.

Standard errors are clustered by county throughout.

Four further specifications are reported. A distributed-lag version adds four quarterly lags of the interaction. A growth version replaces the outcome with the quarterly change in log employment. A pre-trends version adds four leads, and is described below. An event-study version replaces continuous exposure with an indicator for above-median exposure.

The pre-trends test uses the continuous parameterization, not the event-study one. The two disagree about which leads reject, and the continuous version is the one on which this paper's randomization inference was computed. Reporting one test and inferring from the other would be an error, and the choice is stated here so it is not made by accident later.

3.3 Local projections

The second design follows the local-projection approach of Jordà (2005). At each horizon $h = 0, \ldots, 12$ it estimates

$$100 \times \left(\ln E_{c,t+h} - \ln E_{c,t-1}\right) \;=\; \theta_h\left(z_t \times \tilde{s}_c\right) \;+\; \text{(county and quarter means)} \;+\; u_{ct}, \tag{2}$$

where z_t is the policy shock and $\tilde{s}_c$ is exposure standardized to mean zero and unit variance. Both the outcome and the interaction are passed through a two-way within transformation, removing county and quarter means, before estimation without an intercept.

The specification is estimated twice at every horizon. The first pass uses the realized funds rate change as z_t. The second substitutes the Bauer–Swanson surprise. Nothing else changes: same panel, same transformation, same horizons. That is the point of the exercise.

Two features of this design make its coefficients incomparable to equation (1), and both are easy to miss. The outcome is scaled by 100 and cumulated from $t-1$ to $t+h$, and exposure is standardized rather than left in share units. So θ_h is a percentage response per standard deviation of exposure, while β is a log-point response per unit of exposure share. **The two coefficients are not on the same scale and should never be compared directly.** The comparison this paper makes is between two local projections that share a scale, not between the projections and the difference in differences.

Standard errors in the local projections are clustered by quarter rather than by county. This is the appropriate choice: the shock has no cross-sectional variation, so the residual dependence that matters runs across counties within a quarter. It also means inference rests on 72 clusters rather than on the number of observations. Section 5.4 returns to this.

3.4 Randomization inference

The pre-trends result is checked without relying on the clustered variance estimator. Exposure is permuted across counties, the model is re-estimated on the permuted assignment, and the exercise is repeated 200 times per construction. The spread of the placebo estimates is the null distribution, and the reported p-value is the share of permutations whose estimate exceeds the true one in absolute value. The seed is fixed at 20260811.

Two permutation schemes are run. The first shuffles exposure freely across all counties. The second shuffles only within state. The within-state scheme was intended to preserve the geographic concentration of industrial structure that free permutation destroys. It does not do what was intended: restricting the shuffle to same-state counties means exposure can only move to a county with similar exposure, which makes the randomization less aggressive and narrows the null mechanically. Its p-values are therefore anti-conservative and are not quoted as evidence. Both are reported so the reader can see why.

3.5 Sample windows

Every result is reported for two samples: the full 2002 to 2019 panel, and the same panel excluding 2009 through 2012.

The earlier version of this work excluded those years. The zero lower bound is a defensible reason to do so, since the funds rate carries little information about the policy stance when it is pinned near zero. It is not the recorded reason, because no rationale was recorded at the time.

Rather than reconstruct one, both windows are reported throughout. A justification written after seeing the results is worth nothing, and showing both leaves no choice to defend. Where the two disagree, the disagreement is reported. Where they do not, that is stated as well.

3.6 Software and reproducibility

All estimates are produced in R. Fixed-effects models use `fixest`, with `haven` for Stata-format inputs, `sandwich` for clustered variance estimation in the local projections, and `readxl` for the Bauer–Swanson workbook.

An earlier version of the difference-in-differences pipeline was written in Stata. It is retained for provenance only and cannot be executed. Where the two pipelines disagreed, the disagreement was traced rather than reconciled by choosing the preferred answer, and the R implementation is canonical. The resulting reconciliation is documented in full, including the finding that a routine choice about how lagged variables are constructed cut standard errors by 40 to 45 percent without meaningfully moving the point estimates.

That last observation deserves emphasis, because it is this paper's own thesis appearing inside its own code: a defensible-looking construction choice, of a kind no published table would ever display, manufactured precision that was not there.

4 RESULTS

4.1 Where the identifying variation comes from

Figure 1 maps the exposure measure. Exposure is the share of county employment in construction and manufacturing in 2002, taken from County Business Patterns and held fixed thereafter. The share is predetermined, so it cannot respond to later policy.

The spread is wide. Across 3,127 counties the mean share is 0.221 and the median 0.206, with a standard deviation of 0.139. The tenth percentile is 0.043 and the ninetieth is 0.416, a ratio of nearly ten to one. Counties are not mildly more or less exposed than one another. They differ by an order of magnitude.

The map does the work a summary table cannot, because the variation is geographic rather than idiosyncratic. Ranked by mean county exposure, the highest states are Tennessee, South Carolina, Indiana, Alabama and North Carolina, followed by Arkansas, Mississippi, Ohio, Wisconsin and Georgia. That is the southeastern manufacturing belt together with the eastern industrial Midwest. The lowest are the Mountain West and the northern Plains: Montana, North Dakota, Nevada, New Mexico, Arizona, South Dakota, Colorado and Nebraska, along with Alaska and Hawaii, and the District of Columbia lowest of all at 0.020. The District is a single jurisdiction with essentially no manufacturing, and it sits four standard deviations of the state distribution below the exposed South.

That geography matters twice over. It is the source of the cross-sectional variation the design exploits, and it is the reason that variation is spatially correlated, which Section 5.4 returns to.

One feature of the distribution is worth stating rather than leaving to a reader to find. One hundred and forty-three counties, about four and a half percent of the sample, record no construction or manufacturing employment at all in 2002. These are small counties. County Business Patterns suppresses industry detail where publishing it would disclose an individual establishment, so a recorded zero may be a true absence or a suppressed positive value, and the two cannot be distinguished in the published file. Either way those counties enter the estimation at the floor of the exposure distribution. Whether they are influential is not established here.

4.2 Two measures of monetary policy that share almost nothing

Applied work measures monetary policy in two ways. The first is the realized change in the federal funds rate. The second is a high-frequency surprise, here the orthogonalized Bauer–Swanson series. Papers substitute one for the other freely. Figure 2 shows what that substitution does.

Over 2002 to 2019, the two series correlate at -0.010. They share essentially no variation. Some of that is by construction, since the Bauer–Swanson series is purged of macroeconomic and financial information available before the meeting. The point is not that the orthogonality is surprising. The point is that the two measures are treated as interchangeable anyway.

The bottom panels show which of them is contaminated. The realized rate change correlates with contemporaneous national employment growth at $+0.470$ ($p < 0.001$), explaining 22 percent of its variance. The identified surprise correlates at -0.119 ($p = 0.321$), explaining 1.4 percent.

That positive sign is not the effect of policy on employment. If it were, tightening would raise employment, which is the reading this paper rejects. It is the Federal Reserve responding to a strong labor market.

The scatter is visibly leveraged by crisis quarters, so the natural objection is that the correlation is an artifact of 2008. It is not. Excluding 2009 to 2012 the correlation rises to $+0.662$, and the rank correlation over the full window is $+0.502$. Both are reported on the figure.

The reason the correlation strengthens is instructive. At the zero lower bound the funds rate was pinned near zero while employment moved a great deal. Those quarters pair a flat policy series with a volatile cycle, which pulls the correlation toward zero. They are periods when the Federal Reserve could not respond. Removing them does not remove outliers. It removes the episode in which the reaction function was suppressed.

This makes the headline window the conservative one. The endogeneity priced in this paper is larger in the robustness sample than in the sample the main results use.

4.3 The naive estimate is precise, and it points the wrong way

Table 1 reports the shift-share difference-in-differences estimate. Regressing log county employment on the interaction of the quarterly funds rate change with 2002 exposure, absorbing county and quarter fixed effects, gives a contemporaneous coefficient of 0.0332 with a standard error of 0.0066 clustered by county. The p-value is 4×10^{-7}.

Read directly, counties with more construction and manufacturing employment *gain* employment when the Federal Reserve tightens. That contradicts both theory and Carlino and DeFina (1998).

The estimate is also stable across the outcome definition: replacing log employment with employment growth leaves a coefficient of 0.0116 with a standard error of 0.0022. A reader with only this table has a precisely estimated, robust, and wrong-signed result, and no obvious reason to distrust it.

The distributed-lag specification is reported alongside it. Its contemporaneous term is 0.0151 and marginal, and its lagged terms are individually significant at the first and second lags. The coefficients are reported without a characterisation of their profile, for the reason given in Section 5.4. These are separate columns and separate estimands, and they should not be chained into a single sentence.

4.4 First diagnostic: the design fails its own pre-trends test

The design carries its own falsification test. If parallel trends holds, policy changes dated after the outcome should not predict it. Table 2 adds four leads of the interaction to the main specification, and Figure 3 plots them for both sample windows.

They are not zero. Lead one is $+0.0157$ ($p = 0.003$), lead two is $+0.0173$ ($p = 0.002$), and lead three is -0.0217 ($p < 0.001$). Only lead four is indistinguishable from zero. Employment in high-exposure counties was already moving before the policy change arrived.

The verdict does not depend on the sample window. Excluding 2009 to 2012 the same three leads reject, with coefficients that agree to within 0.0015 on every lead. Both windows appear in Figure 3 for that reason.

Three leads reject under clustered inference. The paper reports a count of two throughout, and the next section explains why: only leads two and three also reject under randomization inference. Counting the intersection rather than the union is the conservative choice, and it is the one carried into the abstract.

This is fatal to the causal reading, and it is fatal on the design's own terms. No appeal to an outside benchmark is required.

4.5 The pre-trends failure survives assumption-free inference

A skeptic might answer that the leads are significant only because the clustered standard errors are too small. Figure 4 tests that directly. Exposure is permuted across counties two thousand times, the model is re-estimated on each draw, and the spread of the placebo estimates becomes the null distribution.

The clustered standard errors do not survive this untouched. The ratio of the permutation spread to the clustered standard error is 1.11 for the contemporaneous term and 1.62, 1.41, 1.18 and 0.76 for the four leads. For the first three leads the clustered standard error is too small, by three fifths, by two fifths and by a fifth respectively.

That matters less than it appears, and for a reason worth stating. The lead whose clustered standard error is most understated is lead one, and lead one is the lead that does not reject under permutation. Correcting the understatement pushes it further out of the count, not into it. The two leads that carry the result are the two whose analytic and permutation inference agree.

The leads survive. Lead two rejects at $p = 0.025$ and lead three at $p < 0.0005$ under randomization inference. Lead four does not reject at all. Two of the four pre-period leads therefore reject under both clustered and randomization inference, and that is the count reported throughout.

Lead one is the case the paper declines to resolve. Its permutation p-value is 0.054 against a Monte Carlo standard error of 0.005, so the interval around it still contains the five percent threshold at two thousand draws. Separating it would take roughly eighteen thousand permutations, and the separation would be an artifact of resolution rather than of evidence. It is reported as unsettled. The consequence is that two of four is a lower bound on the number of leads that fail, not a measurement of it, and the design's pre-trends problem is if anything understated here.

Repeating the exercise on the sample that excludes 2009 to 2012 returns the same ratios to two decimal places on three of the four leads, and the same two leads reject. Lead two clears the threshold more narrowly there, at $p = 0.041$. The window choice does not drive this.

4.6 Second diagnostic: precision collapses under an identified shock

The second diagnostic holds the design fixed and changes only the shock. Figure 5 plots local projections of the exposure interaction across horizons zero to twelve, first with the realized rate change and then with the Bauer–Swanson surprise.

With the realized rate change the estimates are positive at all thirteen horizons and significant at eight. With the identified surprise they are negative at all thirteen and significant at none. The same data and the same specification give opposite signs and opposite verdicts.

The excluded window agrees on everything that matters and differs in one detail worth stating plainly. Raw estimates remain positive at thirteen horizons and significant at eight. Identified estimates remain significant at none. They are negative at twelve of thirteen rather than all thirteen, the contemporaneous estimate being indistinguishable from zero. That is not a sign reversal. It is a coefficient the data cannot separate from zero, which is the point being made about every identified estimate in this table.

The sign reversal is not the finding, and the paper does not claim it. Treating the two estimators as independent, which overstates the evidence for a difference because they share a sample, no horizon separates them at five percent. The closest is $p = 0.071$.

What does separate them is precision. Figure 6 plots the two standard-error paths on a common axis. Identified standard errors run from 5.3 to 10.0 times wider than raw ones in the full window, and from 5.8 to 11.7 times wider excluding 2009 to 2012. The gap holds at every horizon in both. The confidence bands do not shift. They inflate.

That is the result. Replacing an endogenous measure of policy with an identified one does not move the estimate to a different place. It reveals that the estimate was never pinned down.

The gap is not produced by the smallest counties. Outcome volatility falls monotonically in county size, so the natural objection is that a minority of very small and very noisy counties manufactures the result. Refitting after dropping the smallest tenth of counties by employment, then the smallest fifth, then the smallest three tenths, moves the ratio range from 5.3–10.0 to 4.5–7.6. No identified estimate becomes significant at any horizon in any of the four samples.

The gap narrows and remains an order of magnitude.

What does erode is the naive estimator's apparent strength. Raw estimates are significant at eight of thirteen horizons in the full sample and at four of thirteen once the smallest three tenths of counties are removed. That is worth conceding plainly: the precision the raw estimator displays is drawn disproportionately from small counties. It is also the paper's point, arriving by a second route. The identified estimator is significant nowhere in any sample, so the comparison the paper rests on is unaffected.

5 DISCUSSION

5.1 What the precision was made of

Section 4.2 showed that the realized rate change carries the business cycle inside it, correlating with contemporaneous employment growth at $+0.470$. Section 4.6 showed that removing that content widens the confidence bands by a factor of five to ten.

Those two facts are the same fact. The naive design looks precise because most of the variation it uses is cyclical, not monetary. Cyclical variation is abundant, systematic, and strongly correlated with the outcome, so it produces tight standard errors. It is also exactly the variation an identification strategy is supposed to remove.

The design was not estimating the effect of monetary policy precisely. It was estimating something else precisely.

5.2 Two diagnostics, two different failures

The paper reports two problems, and they are independent.

The shift-share difference-in-differences fails because parallel trends fails. Its standard errors are not the issue. They survive a mechanism test, a cluster-versus-iid comparison, free-permutation randomization inference, and a within-state permutation. By the last of those they look conservative rather than anti-conservative.

The local projection design fails for the opposite reason. Its point estimates behave sensibly once the shock is identified. What collapses is precision.

A referee unmoved by one still has to answer the other. That is the value of reporting both rather than choosing the more striking one.

5.3 What this means for applied practice

The recommendation is narrow and cheap to follow. Report both estimators.

A shift-share design estimated on realized policy rates should be presented next to the same design estimated on an identified shock, and the comparison should be read on the standard errors rather than the coefficients. A large gap is diagnostic. It says the reported precision is borrowed from variation the design was meant to exclude.

This costs one additional column. It does not require a new dataset, a new estimator, or a structural model.

5.4 Limitations

Five limitations deserve statement rather than burial.

First, exposure-robust inference is untested here. Adão et al. (2019) show that conventional standard errors can be severely understated in shift-share designs when a small number of common sectoral shocks drive exposure. That concern is live in this setting and it is not addressed by the permutation schemes reported above. Free permutation destroys the spatial correlation of exposure, and within-state permutation narrows the null mechanically because industrial structure is regionally correlated. Neither simulates correlated sectoral shocks. The appropriate tool is the Adão–Kolesár–Morales variance estimator, and applying it is left to future work.

Second, the event-study post-period is not identified. Its coefficients move across defensible choices about how lagged variables are constructed, and their significance moves with them. The coefficients are reported. Their shape is not described, because no description of it would be stable.

Third, the identified estimates are imprecise enough that they cannot establish a sign. This paper claims that the naive precision is an artifact. It does not claim to have recovered the true effect. Those are different results, and only the first is supported here.

Fourth, the local projections cluster by quarter, so their inference rests on 72 clusters rather than on the number of county-quarter observations. That is the right choice, because the shock has no cross-sectional variation and the dependence that matters runs across counties within a quarter. It does mean the levels of the standard errors in Figures 5 and 6 reflect a small number of effective clusters.

The comparison this paper makes is unaffected. Both estimators are clustered identically, so the ratio between them cannot be an artifact of the cluster count. The reported ratio of 5 to 10 is a within-design comparison, not a statement about the absolute width of either interval.

Fifth, exposure is not evenly distributed across county sizes, though not in the way a reader would expect. The 2002 exposure share correlates with log median county employment at only +0.16, and the relationship is not monotone. Mean exposure is 0.08 in the smallest tenth of counties, rises to 0.29 in the middle of the size distribution, and falls back to 0.18 in the largest tenth. Construction and manufacturing are concentrated in mid-sized counties, and they are thin in both the rural smallest and the metropolitan largest.

Outcome volatility, by contrast, falls monotonically in size: the standard deviation of the quarterly employment change is 10.6 in the smallest tenth of counties and 2.5 in the largest. The two gradients therefore do not line up, which is the useful part. Exposure is not a proxy for county size, so the interaction is not simply reading a differential response of large places to policy. What remains is narrower: the middle of the size distribution carries most of the identifying variation, and the noisiest counties sit at one end of a gradient that exposure does not follow. That is a reason to check influence, which Section 4.6 does, rather than a reason to distrust the comparison. It does not bear on the precision comparison, which is estimated on a fixed sample with the shock measure as the only thing that changes.

6 CONCLUSION

A shift-share difference-in-differences design applied to county employment returns a precise, robust and wrong-signed estimate of how monetary policy transmits to local labor markets. This paper asked what that precision was made of, and answered that it was made of the business cycle.

Two diagnostics support the conclusion, and they fail in different ways.

The design fails its own pre-trends test. Adding leads of the interaction, two of four reject zero under both clustered and randomization inference, and the verdict holds across sample windows and permutation schemes. Employment in exposed counties was already moving before policy changed, so the estimate cannot be read causally.

Replacing the realized funds rate change with an identified high-frequency surprise does not move the estimate to a different place. It widens the interval by a factor of five to ten in the full window, and six to twelve excluding 2009 to 2012, until no horizon rejects zero in either. The two estimators cannot be told apart at any horizon. The collapse in precision, and not the change in sign, is the finding.

Behind both is a measurement fact. Over this sample the realized rate change correlates with contemporaneous national employment growth at 0.470, while the identified surprise correlates at -0.119. The endogenous measure carries the cycle inside it, and a design built on that measure reports cyclical variation as though it were policy variation.

What this implies for practice

The recommendation is deliberately small, because a recommendation that requires a new estimator will not be followed.

Report the design under both shock measures, and compare them on the standard errors rather than the coefficients. A large gap is diagnostic. It says the reported precision came from variation the identification strategy was meant to remove. This costs one column and no new data.

The wider lesson is about how precision is read. A tight confidence interval is usually taken as evidence that a design is working. In a shift-share setting it can equally be evidence that the design is absorbing the business cycle, and the two cases look identical in a published table. They are told apart by changing the shock, not by inspecting the interval.

Limitations and next steps

Three things this paper does not do.

It does not recover the true employment response. The identified estimates are too imprecise to establish a sign, and claiming one would repeat the error the paper documents.

It does not test exposure-robust inference. Adão et al. (2019) show that conventional standard errors can be badly understated when a small number of common sectoral shocks drive exposure. Two permutation schemes were run and neither addresses that concern: free permutation destroys the spatial correlation of exposure, and within-state permutation narrows the null mechanically. The appropriate instrument is their variance estimator, and applying it is the first item of future work.

It does not describe the shape of the event-study path, in either direction. The post-period coefficients move across defensible constructions of the lagged variables, and the pre-period leads, while stable, are not monotone. The paper reports coefficients and rejects zero. It does not narrate a trajectory that the estimates would not support.

Beyond those, the natural extension is to ask whether the result generalizes. The diagnostic proposed here is not specific to monetary policy or to county employment. Any shift-share design whose shock is a realized policy variable rather than an identified one is exposed to the same problem, and the same one-column check would detect it.

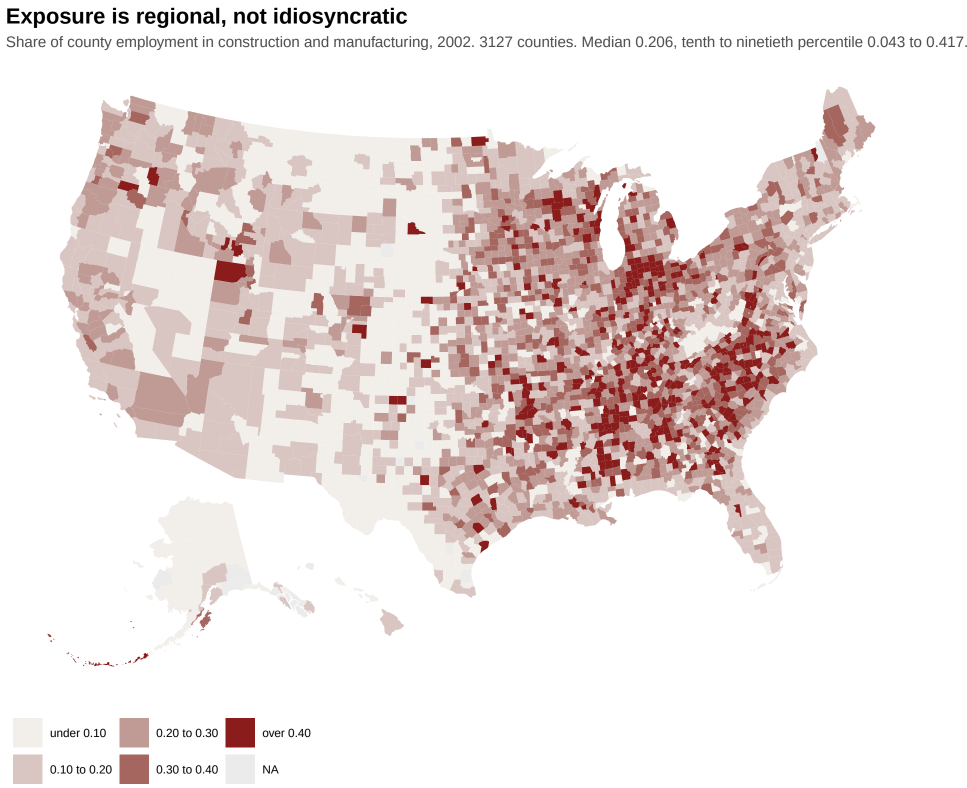

Figure 1: Exposure is regional, not idiosyncratic. Share of county employment in construction and manufacturing, 2002, from County Business Patterns and held fixed thereafter. 3,127 counties. Median 0.206; tenth to ninetieth percentile 0.043 to 0.417. Drawn on 2020 county boundaries: seven counties whose FIPS codes were retired after 2002 have no polygon and are left blank.

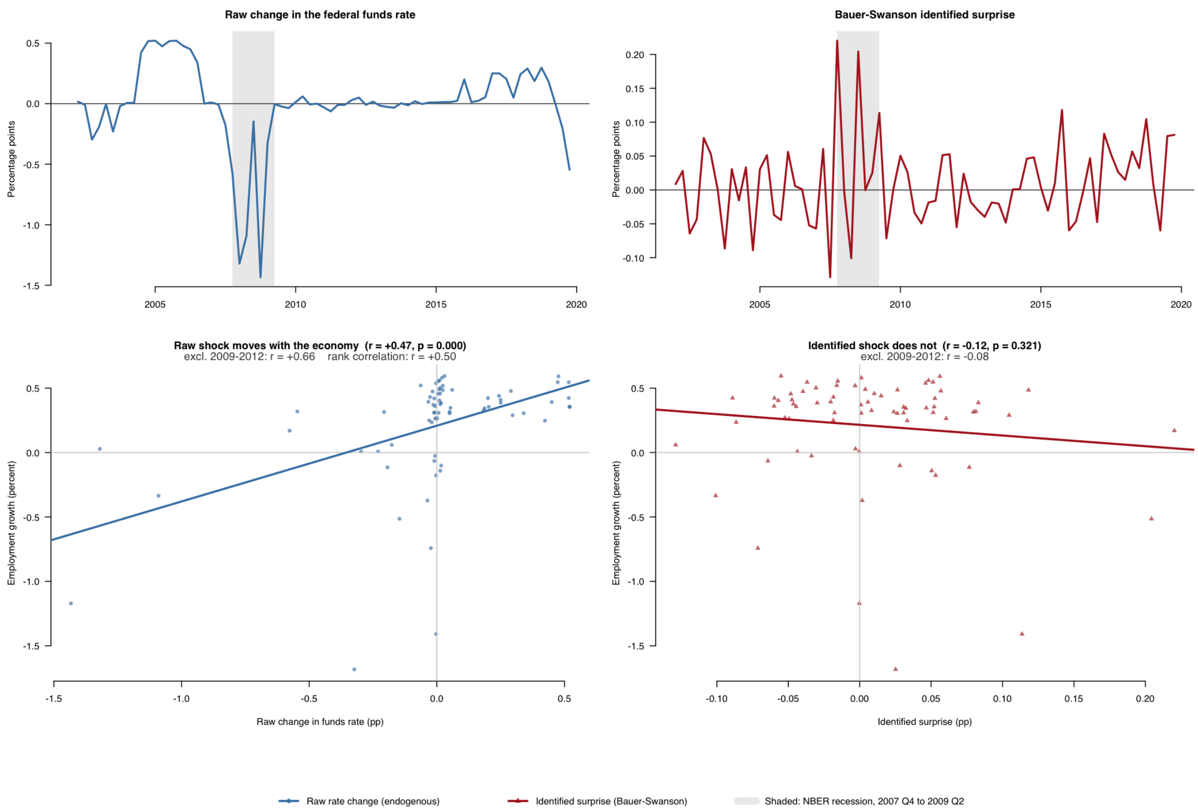

Figure 2: Two measures of monetary policy that share almost nothing. The realized quarterly change in the federal funds rate against the orthogonalized Bauer–Swanson high-frequency surprise, summed to quarters. Both series enter the same design in the sections that follow; nothing else about the specification changes between them.

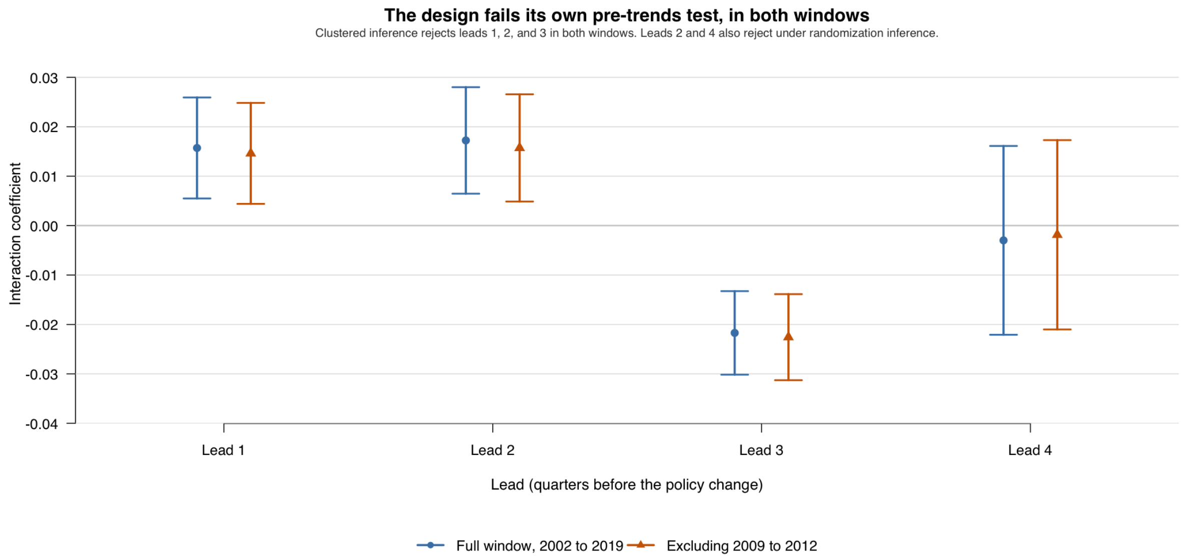

Figure 3: Pre-period leads of the exposure interaction, with 95 percent confidence intervals, standard errors clustered by county. Leads are added to the main specification; a design satisfying parallel trends would place all four at zero. Estimates are shown for the full 2002 to 2019 panel and for the sample excluding 2009 to 2012. Under clustered inference leads one, two and three reject zero at five percent, and the two windows agree to within 0.0015 on every lead. The parallel-trends assumption fails regardless of the window, so the difference-in-differences estimate cannot be read causally.

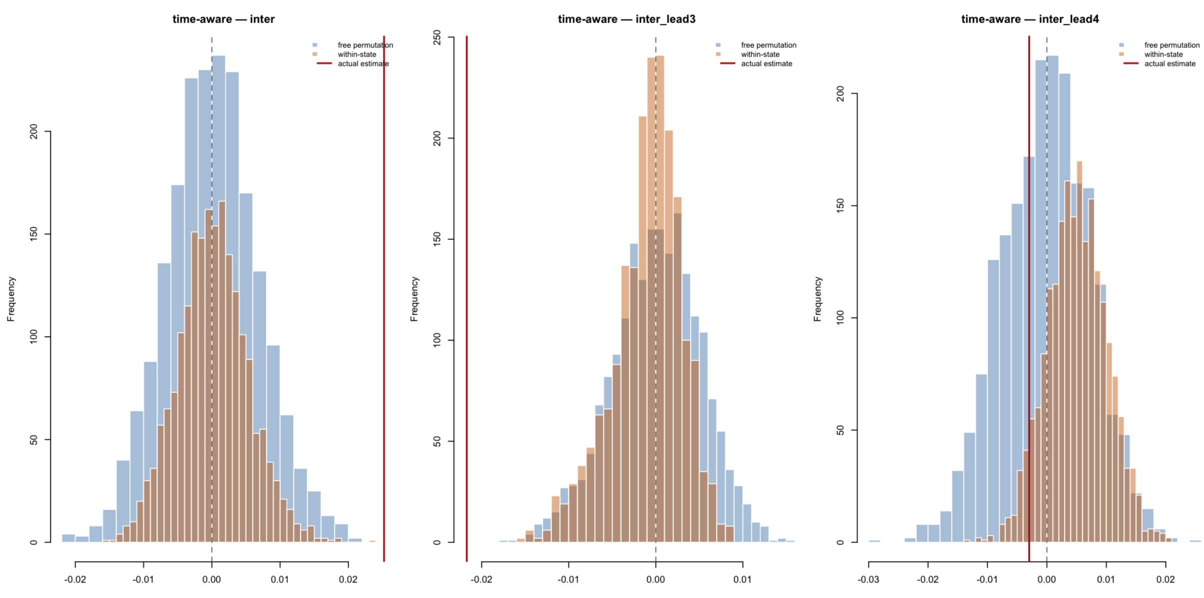

Figure 4: Randomization inference on the pre-period leads. Exposure is permuted across counties two thousand times and the model re-estimated on each draw; the spread of the placebo estimates is the null distribution. The permutation spread runs 1.11 times the clustered standard error for the contemporaneous term and 1.62, 1.41, 1.18 and 0.76 for the four leads, so the clustered errors are too small for the first three leads. Leads two and three reject under permutation, at $p = 0.025$ and $p < 0.0005$. Lead one is reported as unsettled: $p = 0.054$ against a Monte Carlo standard error of 0.005.

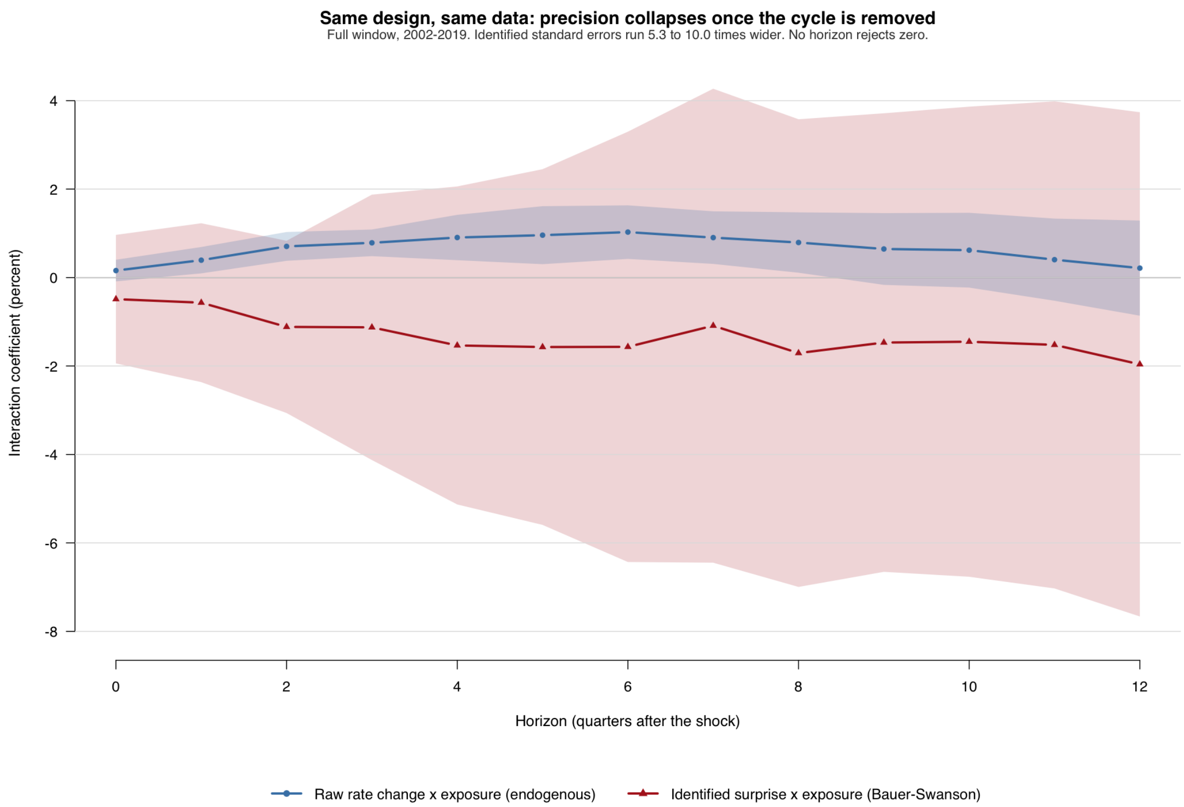

Figure 5: The same design, two shock measures. Local projections of the exposure interaction across horizons zero to twelve, both sample windows. The raw estimator uses the quarterly change in the federal funds rate; the identified estimator substitutes Bauer–Swanson high-frequency surprises, holding the design fixed. Raw estimates are positive at all thirteen horizons and significant at eight; identified estimates are negative at all thirteen and significant at none. The two estimators are not statistically distinguishable at any horizon, the closest being $p = 0.071$ at $h = 2$, so the figure does not establish a change in sign. It establishes a collapse in precision.

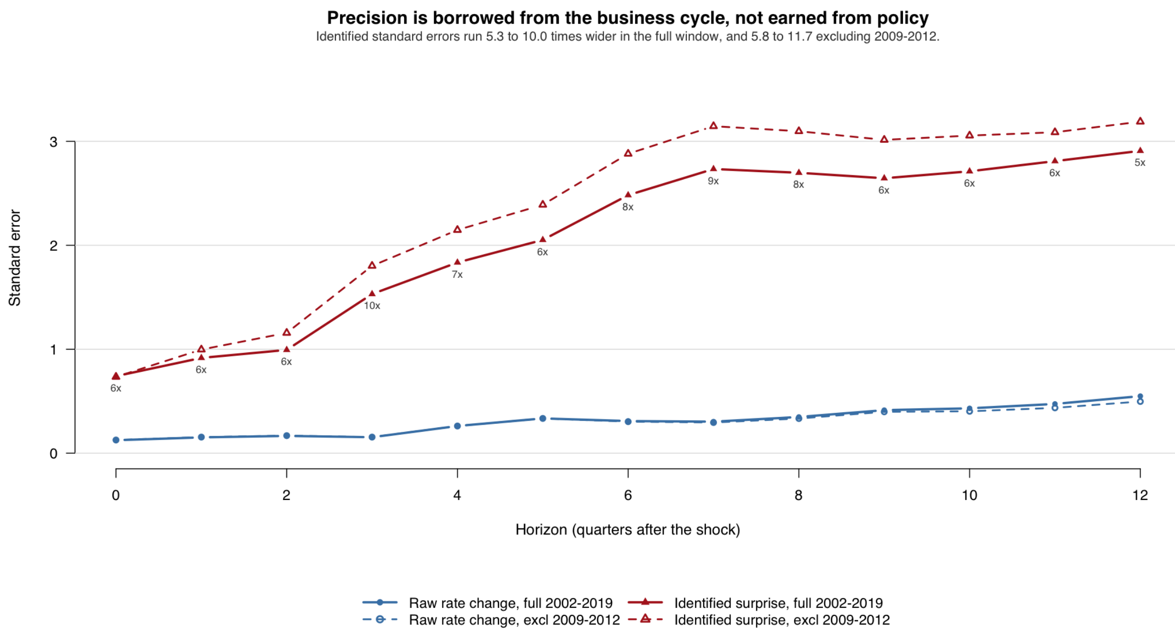

Figure 6: Precision, not sign, is what changes. Standard errors of the local projection estimates by horizon, both sample windows. Colour distinguishes the shock measure; line type distinguishes the window. Identified standard errors run 5.3 to 10.0 times wider than raw ones over 2002–2019, and 5.8 to 11.7 times wider excluding 2009–2012. The gap holds at every horizon in both windows. Point estimates are in Figure 5.

Table 1: Shift-share difference-in-differences estimates of the exposure interaction. The outcome is log county employment. Exposure is the 2002 County Business Patterns share of county employment in construction and manufacturing, held fixed. Standard errors, clustered by county, in parentheses. 3,127 counties. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

	Full panel 2002–2019 (1)	Excluding 2009–2012 (2)
Exposure $\times$ Δ federal funds rate	0.0332***	0.0318***
	(0.0066)	(0.0064)
County fixed effects	Yes	Yes
Quarter fixed effects	Yes	Yes
Observations	224,952	174,968

The two estimates differ by 0.0014, about a fifth of a standard error, so the result is not produced by the zero-lower-bound period. Section 4.6 shows that this precision does not survive an identified shock.

Table 2: Pre-trends test. Four leads of the exposure interaction are added to the main specification. Under parallel trends all four are zero. Standard errors, clustered by county, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

	Full panel 2002–2019 (1)	Excluding 2009–2012 (2)
Exposure $\times$ Δ federal funds rate	0.0252***	0.0229***
	(0.0060)	(0.0057)
lead 1	0.0157***	0.0146***
	(0.0052)	(0.0052)
lead 2	0.0172***	0.0157***
	(0.0055)	(0.0055)
lead 3	-0.0217***	-0.0226***
	(0.0043)	(0.0044)
lead 4	-0.0030	-0.0019
	(0.0097)	(0.0098)
County fixed effects	Yes	Yes
Quarter fixed effects	Yes	Yes
Observations	212,468	162,484

Leads one, two and three reject zero at five percent in both windows, so the parallel-trends assumption fails and the difference-in-differences estimate cannot be read causally. The paper reports a count of two, the leads that also reject under randomization inference; see Section 4.4.

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